Time series analysis — exercise sheet 2

AB–204 Arctic population ecology, October 28, 2026

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This exercise sheet is about the basics of linear regression (Lecture 2).

Before you start: open your script from exercise sheet 1, and run the lines that import the data and create the new variables. You need:

```
library(viridisLite) # colours for categories
bikes <- read.csv("bike_sharing.csv") # all days, 2016-2026
bikes$windspeed_kmh <- bikes$windspeed * 1.609 # wind speed in km/h
bikes$temperature_c <- (bikes$min_temperature - 32) * 5 / 9 # min. temp. in C
```

Use the **full** data set `bikes` for this exercise sheet.

Exercise 1: simple linear regression

1. Use the `lm()` function to fit the simple linear model

$$\mathbb{E}(\text{number of bikes}_t) = \beta_0 + \beta_1 \cdot \text{temperature}_t$$

   to the bike sharing data, and assign the fitted model to the object `mod_1`.
2. Use the `summary()` function to print the model output.
3. What is the interpretation of the estimated coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$? Is the effect of temperature statistically significant? How much of the variability in the number of bikes is explained by temperature?
4. Create a scatter plot of `temperature_c` against `count`, and add the fitted regression line.
5. Perform residual checks: plot a histogram of the residuals with a normal density, a QQ plot, and the residuals against temperature. Do the residuals follow a normal distribution? Do you see any patterns?

Exercise 2: multiple linear regression and predictions

1. Create a new variable `rainy` that is 0 on dry days (`precipitation == 0`) and 1 on rainy days (`precipitation > 0`).
2. Use the `lm()` function to fit the multiple linear model

$$\mathbb{E}(\text{number of bikes}_t) = \beta_0 + \beta_1 \cdot \text{temperature}_t + \beta_2 \cdot \text{wind speed}_t + \beta_3 \cdot \text{rainy}_t$$

   to the bike sharing data, and assign the fitted model to the object `mod_2`.
3. Use the `summary()` function to print the model output.
4. What are the interpretations of the estimated coefficients $\hat{\beta}_0, \ldots, \hat{\beta}_3$? Are the effects of temperature, wind speed, and rain statistically significant?
5. Create a 3-dimensional scatter plot of `temperature_c` against `windspeed_kmh` against `count`, with different colours for rainy and dry days (see exercise sheet 1, Exercise 2.4).
6. Add two regression surfaces to the 3-dimensional scatter plot from 2.5, one for dry days and one for rainy days. How do they differ?
   Hint: use `surface()` as on the slides, with `rainy = 0` and `rainy = 1` in `predict()`.
7. Perform residual checks as in 1.5. Do the residuals follow a normal distribution?
8. Use the `AIC()` function to determine whether the simple or the multiple linear model is better.
   Hint: the AIC can only compare models that were fitted to the **same** observations — `mod_2` leaves out the days without wind speed, `mod_1` doesn't.
9. Use `mod_2` and the `predict()` function to predict the number of bikes on a dry and on a rainy day with a minimum temperature of 20 °C and a wind speed of 10 km/h.

Questions? menti.com/al94rhvnrre9.

10. Now predict the number of bikes on a rainy day with $-10\,°$C and a wind speed of 40 km/h. What do you notice? What does this tell you about the model?
11. **Bonus:** Can you find an even better model using other covariates?